

Sara Hamzeloui

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Research Profile

Postdoctoral Researcher working on design, fabrication, and characterization of III–V semiconductor devices. Research interests include high-frequency circuit design, semiconductor process development, and advanced device characterization. Extensive experience across the complete technology development cycle, from cleanroom process integration and fabrication of indium phosphide (InP) heterojunction bipolar transistors (HBTs) to compact modeling of active and passive devices, MMIC process design kit (PDK) development, and comprehensive small- and large-signal characterization of compound semiconductor technologies.

Education

- **Postdoctoral Researcher** Zürich, Switzerland (Dec 2024 – Present)
ETH Zurich, Department of Electrical Engineering and Information Technology
Millimeter-Wave Electronics (MWE) Group
Research Topic: Design and Implementation of High-Speed Integrated Circuits Using InP/GaAsSb Double Heterojunction Bipolar Transistors (DHBTs) MMIC Technology for Beyond 300 Gb/s Optical Communications
- **Doctor of Sciences (Ph.D.)** Zürich, Switzerland (Jan 2019 – Nov 2024)
ETH Zurich, Department of Electrical Engineering and Information Technology
Millimeter-Wave Electronics (MWE) Group
Thesis Topic: Design, Fabrication, and Characterization of High-Power InP/GaAsSb Double Heterojunction Bipolar Transistors (DHBTs) for mm-Wave Power Amplifiers
- **Master of Science (M.Sc)** Tehran, Iran (Sep 2016 – Sep 2018)
Sharif University of Technology, Department of Electrical Engineering
Thesis Topic: Design and Implementation of a 28 GHz Phased-Array Receiver Front-End in the 180-nm CMOS Technology for 5G Applications
- **Bachelor of Science (B.Sc)** Tehran, Iran (Sep 2011 – Sep 2015)
Sharif University of Technology, Department of Electrical Engineering

Professional Experience

- **Device and MMIC Fabrication**
 - Skilled in III–V device fabrication using optical and e-beam lithography, ICP/RIE dry etching, ALD/PECVD deposition, and SEM/AFM characterization.
 - Developed an industry-compatible InP/GaAsSb DHBT MMIC fabrication process achieving high yield for complex circuits at the Millimeter-Wave Electronics Group, ETH Zurich.
 - Technical lead and tool owner for ICP, ALD, and EVG mask aligner systems in the FIRST-ETHZ cleanroom, responsible for operation, maintenance, and user training.
- **High-Frequency Circuit Design**
 - Designed high-power, high-efficiency W- and G-band power amplifiers (PAs) using InP HBT technology for mm-wave applications.
 - Designed InP HBT-based analog multiplexers (AMUX) enabling beyond 300 Gb/s data rates for high-speed optical communication systems.
 - Designed low-noise amplifiers (LNAs) and phase shifters (PSs) in 180-nm CMOS technology for 5G RF front-end communication systems.
- **Device and MMIC Characterization**

- Set up, calibrated, and operated DC/RF on-wafer characterization systems for device and circuit measurements up to 220 GHz.
- Performed large-signal load-pull measurements up to 170 GHz and linearity characterization (IIP3/OIP3) up to 50 GHz.
- Developed a complete MMIC process design kit (PDK) for InP/GaAsSb DHBT technology based on broadband measurements up to 170 GHz.
- Collaborated with researchers at TU Dresden to develop and validate a HICUM/L2 large-signal compact model for InP/GaAsSb DHBT devices.

• Teaching and Educational Experience

- Served as Teaching Assistant and Session Lecturer for Semiconductor Devices, High-Frequency Design Techniques, and High-Speed Signal Propagation at ETH Zurich (2019–2021).
- Supervised and mentored multiple Master’s and Bachelor’s students on RF, semiconductor device, and circuit design research projects.
- Reviewed manuscripts for IEEE journals including IEEE Microwave and Wireless Technology Letters and IEEE Transactions on Microwave Theory and Techniques.

Skills

- **Semiconductor Processing:** III-V Device Fabrication (InP HBTs/UTCs), Lithography, Etching, Metallization.
- **Modeling and PDK:** Compact modeling, AEL PDK development, DRC automation.
- **Circuit Design:** LNA, PA, Phase-shifter, AMUX.
- **Measurement and Characterization:** S-parameters, Load-pull, Linearity, Noise.
- **Software:** Keysight ADS (Schematic/EM/RFpro), Ansys HFSS, Cadence, MATLAB.

Awards and Honors

- ETH Medal Recipient for Outstanding Doctoral Thesis, 2025. [link](#)
- Swiss National Science Foundation (SNSF) Postdoctoral Mobility Fellowship Recipient, 2024.
- 2024 Global Symposium on mm-Wave and Terahertz (GSMM) Best Student Paper Award for “94 GHz Two-Way-Combined Power Amplifiers with 14.2 dBm Peak Output Power in 300-nm InP/GaAsSb DHBT Technology”, First Author. [link](#)
- 2022 *IEEE* EDS Paul Rappaport Award for Best *IEEE* Transactions on Electron Devices (TED) Paper “InP/GaAsSb Double Heterojunction Bipolar Transistor Emitter-Fin Technology With $f_{\text{MAX}} = 1.2$ THz”, Co-author. [link](#)
- 2023 *IEEE* BiCMOS and Compound Integrated Circuits and Technology Conference (BCICTS) Best Student Paper Award for “Thermal Characterization of InP/GaAsSb DHBTs: Effect of Emitter and Collector Layers”, Co-author.
- 2023 *IEEE* BiCMOS and Compound Integrated Circuits and Technology Conference (BCICTS) Finalist for Best Student Paper Award for “Multi-Finger 250-nm InP/GaAsSb DHBTs with Record 37.3% Class-A PAE at 94 GHz”, First Author.
- Best Teaching Assistant Award for teaching “High-Frequency Design Techniques” in the Department of Electrical Engineering and Information Technology, ETH Zurich, December 2021. [link](#)

Publication List

Journals

1. **Sara Hamzeloui**, Akshay M Arabhavi, Filippo Ciabattini, Ralf Flückiger, Diego Marti, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo R Bolognesi. “High Power InP/Ga(In)AsSb DHBTs for Millimeter-Wave PAs: 14.5 dBm Output Power and 10.4 mW/ μm^2 Power Density at 94 GHz”, **IEEE Journal of Microwaves**, **2022**.
2. **Sara Hamzeloui**, Akshay M Arabhavi, Filippo Ciabattini, Giorgio Bonomo, Mojtaba Ebrahimi, Rimjhim Chaudhary, Markus Müller, Olivier Ostinelli, Michael Schröter, Colombo R Bolognesi. “Record 35% Power-Added Efficiency at 170 GHz in 300-nm InP/GaAsSb DHBTs”, **IEEE Microwave and Wireless Technology Letters**, **2024**.
3. **Sara Hamzeloui**, Akshay M Arabhavi, Filippo Ciabattini, Mojtaba Ebrahimi, Giorgio Bonomo, Andrea Cercaci, Olivier Ostinelli, Colombo R Bolognesi. “94-GHz Four-Way Combined Power Amplifiers With 19.2-dBm Output Power in 250-nm InP/GaAsSb DHBT Technology”, **IEEE Microwave and Wireless Technology Letters**, **2025**.

4. **Sara Hamzeloui**, Akshay M Arabhavi, Filippo Ciabattini, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo R Bolognesi, “Achieving Optimal Impact Ionization-Limited Breakdown Voltages in Scaled mm-Wave InP/GaInAsSb DHBTs”, **IEEE Transactions on Electron Devices**, **2025**.
5. Filippo Ciabattini, **Sara Hamzeloui**, Akshay M Arabhavi, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo Bolognesi, “Large-signal characterization of THz emitter-fin InP/GaAsSb DHBTs with record G-band power added efficiency”, **International Journal of Microwave and Wireless Technologies**, **2025**.
6. Akshay M Arabhavi, Filippo Ciabattini, **Sara Hamzeloui**, Ralf Flückiger, Tamara Saranovac, Daxin Han, Diego Marti, Giorgio Bonomo, Rimjhim Chaudhary, Olivier Ostinelli, Colombo R Bolognesi, “InP/GaAsSb Double Heterojunction Bipolar Transistor Emitter-Fin Technology With $f_{MAX} = 1.2$ THz”, **IEEE Transactions on Electron Devices**, **2022**.
7. Akshay M Arabhavi, Rimjhim Chaudhary, Ralf Flückiger, Diego Marti, **Sara Hamzeloui**, Filippo Ciabattini, Wei Quan, Martin Leich, Olivier Ostinelli, Colombo R Bolognesi, “Type-II GaInAsSb/InP Uniform Absorber High Speed Uni-Traveling Carrier Photodiodes”, **Journal of Lightwave Technology**, **2021**.
8. Rimjhim Chaudhary, Akshay M Arabhavi, Laurenz Kulmer, **Sara Hamzeloui**, Martin Leich, Olivier Ostinelli, Juerg Leuthold, Colombo R Bolognesi, “Wideband Type-II GaInAsSb/InP Uni-Traveling Carrier Photodiodes for Near 300 Gbps Communications”, **Journal of Lightwave Technology**, **2023**.
9. Viktor Kazantsev, Markus Mueller, Hendrik Leenders, Josephine Faisst, Xiaodi Jin, **Sara Hamzeloui**, Colombo R Bolognesi, Christoph Jungemann, Michael Schroeter, “Analysis of Multi-Valley Phenomena in InP/GaAsSb DHBTs using the Boltzmann Transport Equation”, **IEEE Transactions on Electron Devices**, **2025**.
10. Marina Deng, Chhandak Mukherjee, Lucas Reveil, Akshay M Arabhavi, **Sara Hamzeloui**, Colombo Bolognesi, Magali De Matos, Cristell Maneux, “InP/GaAsSb double heterojunction bipolar transistor characterization and compact modeling up to 500 GHz”, **IEEE Transactions on Electron Devices**, **2024**.

Conferences

11. **Sara Hamzeloui**, Filippo Ciabattini, Akshay M Arabhavi, Wei Quan, Diego Marti, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo R Bolognesi, “Highly-Linear InP/GaAsSb DHBTs with 18.2 dBm OIP3 at 45 GHz: Comparison of Common-Base and Common-Emitter Structures”, **IEEE BCICTS**, **2022**.
12. **Sara Hamzeloui**, Akshay M Arabhavi, Filippo Ciabattini, Mojtaba Ebrahimi, Markus Müller, Olivier Ostinelli, Michael Schröter, Colombo R Bolognesi, “Multi-Finger 250-nm InP/GaAsSb DHBTs with Record 37.3% Class-A PAE at 94 GHz”, **IEEE BCICTS** **2023**.
13. **Sara Hamzeloui**, Akshay M Arabhavi, Filippo Ciabattini, Diego Marti, Ralf Flückiger, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo R Bolognesi, “Demonstration of 300-nm InP/GaInAsSb DHBT MMIC Technology in a 60-160GHz Ultra-Broadband Amplifier Test Vehicle”, **EuMIC**, **2022**.
14. **Sara Hamzeloui**, Mojtaba Ebrahimi, Akshay M. Arabhavi, Filippo Ciabattini, Olivier Ostinelli, and Colombo R. Bolognesi, “Multi-Finger InP/GaAsSb DHBTs with Record 15 dBm Pout,sat and 30 % Power-Added Efficiency at 94 GHz”, **CSW**, **2023**.
15. **Sara Hamzeloui**, Akshay M. Arabhavi, Olivier Ostinelli and Colombo R. Bolognesi, “Optimal Impact Ionization-Limited Breakdown Voltages in mm-Wave InP/GaInAsSb DHBTs”, **CSW**, **2023**.
16. **Sara Hamzeloui**, Akshay M. Arabhavi, Filippo Ciabattini, Giorgio Bonomo, Mojtaba Ebrahimi, Olivier Ostinelli and Colombo R. Bolognesi, “94 GHz Two-Way-Combined Power Amplifiers with 14.2 dBm Peak Output Power in 300-nm InP/GaAsSb DHBT Technology”, **GSMM**, **2024**.
17. Andrea Cercaci, **Sara Hamzeloui**, Filippo Ciabattini, Mojtaba Ebrahimi, Akshay M. Arabhavi, Olivier Ostinelli and Colombo R. Bolognesi, “Multi-Finger InP/GaAsSb DHBTs with Non-Uniform Emitter Finger Widths for Improved Thermal Performance”, **IEEE BCICTS**, **2025**.
18. Wei Quan, Akshay M Arabhavi, Diego Marti, **Sara Hamzeloui**, Olivier Ostinelli, Colombo R Bolognesi, “InP/GaAsSb DHBT Power Performance with 30% Class-A PAE at 94 GHz”, **IEEE BCICTS**, **2019**.
19. Colombo R Bolognesi, Akshay M Arabhavi, Romain Hersent, **Sara Hamzeloui**, Filipe Jorge, Xin Wen, Muriel Riet, Mathieu Luisier, Virginie Nodjiadjim, Filippo Ciabattini, Colin Mismar, Olivier Ostinelli, Agnieszka Konczykowska, “InP/GaAsSb DHBTs: THz Analog Performance and Record 180-Gb/s 5.5Vppd-Swing PAM-4 DAC-Driver”, **IEEE BCICTS**, **2022**.

20. Akshay M Arabhavi, Filippo Ciabattini, **Sara Hamzeloui**, Ralf Flückiger, Tamara Popovic, Daxin Han, Diego Marti, Giorgio Bonomo, Rimjhim Chaudhary, Olivier Ostinelli, Colombo R Bolognesi, “InP/GaAsSb Double Heterojunction Bipolar Transistor Technology with $f_{\text{MAX}} = 1.2$ THz”, **IEEE IEDM**, **2021**.
21. Rimjhim Chaudhary, Akshay M Arabhavi, Laurenz Kulmer, **Sara Hamzeloui**, Marco Eppenberger, Martin Leich, Olivier Ostinelli, Juerg Leuthold, Colombo R Bolognesi, “High-Power Performance of Type-II GaInAsSb/InP Uniform Absorber Uni-Traveling Carrier Photodiodes”, **OFC**, **2023**.
22. Wei Quan, **Sara Hamzeloui**, Akshay M Arabhavi, Ralf Flückiger, Olivier Ostinelli, CR Bolognesi, “140-190 GHz Broadband Amplifier in 300-nm InP/GaAsSb DHBT Technology”, **EuMC**, **2020**.
23. Filippo Ciabattini, Akshay M Arabhavi, **Sara Hamzeloui**, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo R Bolognesi, “Thermal Characterization of InP/GaAsSb DHBTs: Effect of Emitter and Collector Layers”, **IEEE BCICTS**, **2023**.
24. Markus Müller, Mario Krattenmacher, Hendrik Leenders, **Sara Hamzeloui**, Colombo R Bolognesi, Christoph Jungemann, Michael Schröter, “Physics-Based Compact Modeling of the Transfer Current in III-V DHBTs with the Generalized Integral Charge Control Relation”, **IEEE BCICTS**, **2023**.
25. Filippo Ciabattini, **Sara Hamzeloui**, Wei Quan, Akshay M. Arabhavi, Mojtaba Ebrahimi, Diego Marti, Olivier Ostinelli, Colombo R Bolognesi, “InP/GaAsSb DHBT Linearity Performance Versus Device Geometry”, **CSW**, **2023**.
26. Akshay M Arabhavi, Marina Deng, Filippo Ciabattini, **Sara Hamzeloui**, Tamara Saranovac, Rimjhim Chaudhary, Mojtaba Ebrahimi, Olivier Ostinelli, Cristell Maneux, Colombo R Bolognesi, “THz InP/GaAsSb DHBTs with Record $f_{\text{AVG}} = 800$ GHz: Characterization to 330 GHz”, **IEEE IEDM**, **2023**.
27. Akshay M. Arabhavi, **Sara Hamzeloui**, Filippo Ciabattini, Olivier Ostinelli, and Colombo R. Bolognesi, “Terahertz InP/GaAsSb Double Heterojunction Bipolar Transistors”, **SSDM**, **2022**.
28. Akshay M. Arabhavi, Filippo Ciabattini, **Sara Hamzeloui**, Ralf Flückiger, Tamara Saranovac, Daxin Han, Diego Marti, Giorgio Bonomo, Rimjhim Chaudhury, Olivier Ostinelli, and Colombo R. Bolognesi, “THz InP/GaAsSb DHBT Characterization from 300 to 7 K”, **WOCSICE**, **2022**.
29. Colombo Bolognesi, Filippo Ciabattini, **Sara Hamzeloui**, Akshay M Arabhavi, Mojtaba Ebrahimi, Andrea Cercaci, Giorgio Bonomo, Olivier Ostinelli, “Recent Developments in Type-II InP/Ga(In)AsSb DHBTs: Device Physics, Performance and New Perspectives”, **SSDM**, **2025**.
30. Colombo Bolognesi, Rimjhim Chaudhary, Akshay M Arabhavi, Olivier Ostinelli, **Sara Hamzeloui**, Martin Leich, “Development of High-Performance 1550 nm GaInAsSb Unitraveling Carrier Photodiodes”, **IEEE Photonics Conference**, **2024**.
31. Filippo Ciabattini, Akshay M Arabhavi, **Sara Hamzeloui**, Giorgio Bonomo, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo Bolognesi, “High-Temperature Stability Base Ohmic Contacts for InP/GaAsSb DHBTs”, **IEEE BCICTS**, **2024**.
32. Filippo Ciabattini, **Sara Hamzeloui**, Akshay M Arabhavi, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo Bolognesi, “G-Band Large-Signal Characterization of InP/GaAsSb DHBTs with Record 38% Power Added Efficiency at 170 GHz”, **EuMIC**, **2024**.
33. Rimjhim Chaudhary, Akshay M Arabhavi, **Sara Hamzeloui**, Martin Leich, Olivier Ostinelli, Colombo Bolognesi, “Type-II GaInAsSb/InP Modified Uni-Traveling Carrier Photodiodes Under Zero-Bias Operation”, **OFC**, **2024**.
34. Filippo Ciabattini, **Sara Hamzeloui**, Akshay M Arabhavi, Mojtaba Ebrahimi, Olivier Ostinelli, Colombo Bolognesi, “G-Band Load-Pull Characterization of High-Efficiency Emitter-Fin InP/GaAsSb DHBTs”, **GSMM**, **2024**.
35. Akshay M Arabhavi, **Sara Hamzeloui**, Wei Quan, Filippo Ciabattini, Olivier Ostinelli, Colombo Bolognesi, “InP/GaInP Composite-Collector for Improved Breakdown Voltage in the InP/GaAsSb DHBTs”, **IEEE EDTM**, **2024**.

Book Chapter

36. Colombo R Bolognesi, Akshay M Arabhavi, **Sara Hamzeloui**, “High-Speed InP-Based heterojunction bipolar transistors”, Reference Module in Materials Science and Materials Engineering, Elsevier, 2023, ISBN 9780128035818.